MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

STUDENT:
Jules STEVENOT

SUPERVISOR:
Bruno BELLOMO

January-June 2026

Deliberately left blank

ACKNOWLEDGEMENTS

The author expresses deep gratitude to Bruno Bellomo for his guidance and support throughout this research. Bruno, as a mentor, constantly took the time to explore ideas, ensure our understanding, and seek details. His expertise in Quantum Mechanics has been fundamental to the successful completion of this Master's thesis. I am grateful to conclude my short career as a physicist with such a demanding journey, thanks to Bruno.

I would like to thank the members of the jury for their time and interest in this work, as well as for the opportunity to present my work in front of such esteemed peers.

I also would like to thank my colleagues, of a few months, for their insightful discussions and support during this internship: Silvio Da Silva, Jana El Badawi, Anjana Mohan, and Léo Jacquerot-Legros.

Finally, I would like to thank my friends and family for their never-ending support: Erwan Le Dœuff, Titouan Alphonse, and Nabyan Arkan for these two years of Master's studies and friendship together, also, thank to Maxence Bonin and Vincent Girod my brothers from other mothers. I owe everything to my parents, Catherine and Lionel, for my life and education, and, to my brother Pierre for his resilience and skills. Lastly, I want to thank my other half, Alexiane.

L'auteur exprime sa profond gratitude envers Bruno Bellomo pour ses conseils et son soutien tout au long de ce travail de recherche. Bruno, en tant que mentor, a constamment pris le temps d'explorer des idées, de s'assurer de notre compréhension et d'approfondir les détails. Son expertise en mécanique quantique a été fondamentale dans la bonne conduite de ce mémoire de master. Je suis reconnaissant de pouvoir conclure ma courte carrière de physicien par un parcous aussi exigeant, grâce à Bruno.

Je tiens à remercier les membres du jury pour le temps et l'intérêt qu'ils porteront à ce travail, ainsi que pour l'opportunité de le présenter devant des pairs reconnus.

Je souhaite tout de même remercier mes collègues de quelques mois pour leurs discussions enrichissantes et leur soutien durant ce stage: Silvio Da Silva, Jana El Badawi, Anjana Mohan et Léo Jacquerot-Legros.

Enfin, je tiens à remercier mes amis et ma famille pour leur soutien indéfectible : Erwan Le Dœuff, Titouan Alphonse et Nabyan Arkan pour ces deux années d'études de Master et amitié partagées, mais également, merci à Maxence Bonin et Vincent Girod mes frères d'autre mères. Je dois tout à mes parents, Catherine et Lionel, pour ma vie et mes études, ainsi qu'à mon frère Pierre qui a su être résilient et talentueux. Pour finir, je remercie ma meilleure moitié et compagne, Alexiane.

« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

CONTENTS

Introduction **1**

I From closed systems to continuous measurements **2**

I.i) Quantum mechanics 2

I.i.1⟩ States and time evolution 2

I.i.2⟩ Observables 2

I.i.3⟩ Purity of states 3

I.i.4⟩ Postulates of quantum mechanics 4

I.ii) Open quantum systems 5

I.iii) Quantum measurement theory 5

II Continuously measured quantum systems **6**

A Representations of quantum mechanics **7**

A.a) Schrödinger and Heisenberg pictures 7

A.b) Interaction picture 7

CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in space A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

I From closed systems to continuous measurements — in a nutshell

In this section we will briefly summarize the definitions needed to our study.

I.i) Generalities on quantum mechanics

I.i.1⟩ States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H} \tag{1}$$

The evolution of a system depends on its energy, the Hamiltonian H is an hermitian[2] operator that accounts for the total energy of a system. The Schrödinger equation states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle \tag{2}$$

Instead of looking at the state at time t we can use an unitary[3] operator $\hat{U}$ such that, $|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$. From now on, we will omit the $\hat{}$ symbol over operators when it is obvious. We get,

$$\dot{U}(t,t_0) = -\frac{i}{\hbar} H(t) U(t,t_0) \tag{3}$$

One can solve this differential equation in general by using a chronological time-ordering operator [3]. Giving for the state evolution,

$$|\psi(t)\rangle = \mathcal{T}_{\leftarrow} \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} H(s)\mathrm{d}s\right]|\psi(t_0)\rangle \tag{4}$$

Or, when the Hamiltonian is time-independent,

$$|\psi(t)\rangle = \underbrace{e^{-\frac{iH(t-t_0)}{\hbar}}}_{U(t,t_0)}|\psi(t_0)\rangle \tag{5}$$

This evolution of the state with respect to time is said **unitary and reversible** because U is unitary, and, because this equation also describe the time reversed evolution $t \to t_0$: $|\psi(t_0)\rangle = U^\dagger(t,t_0)\,|\psi(t)\rangle$.

I.i.2⟩ Observables

Any physical quantity is represented in quantum mechanics by an **observable**: an operator whose eigenvectors form an orthonormal basis of the Hilbert space $\mathcal{H}$ [2] and whose eigenvalues are positive to represent a physical measurement. For finite dimension Hilbert space, hermitian operators[4] are observables. In the following we will consider finite dimension, non-degenerate cases to illustrate but one can generalize [2].

[2] $\hat{H}^\dagger = \hat{H}$, where † is the conjugate transpose operation, defining the hermitian adjoint.

[3] $UU^\dagger = U^\dagger U = \mathbb{I}$, where $\mathbb{I}$ is the identity.

[4] Hermitian operators always form a basis in finite dimension, we can use Gram-Schmidt process to find its orthonormal counterpart and being self-adjoint implies having real eigenvalues

A natural operator whose eigenbasis is particularly interesting is the Hamiltonian, its eigenvalues correspond to the possible energy levels of a system.
Let us denote $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document.

$$\forall n \in I_{\mathcal{H}}, \quad H\left|\phi_n\right\rangle = E_n\left|\phi_n\right\rangle \tag{6}$$

Let us call $\{\phi_n\}_{n\in I_{\mathcal{H}}}$ the family of orthonormal eigenvectors of H which forms a basis of $\mathcal{H}$.
Since it forms a basis of the Hilbert space, one can describe any state in this very basis:

$$\forall \left|\psi\right\rangle \in \mathcal{H}, \qquad \left|\psi\right\rangle = \sum_{n\in I_{\mathcal{H}}} \underbrace{c_n}_{\langle\phi_n|\psi\rangle} \left|\phi_n\right\rangle \tag{7}$$

This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian. We can complete this part with the normalization of the state, for proper handling of scalar products, we need to set $|\psi| = \sqrt{\langle\psi|\psi\rangle} = 1$. The condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$ is thus equivalent.

I.i.3⟩ Purity of states

Taking the qubit example, let $\alpha, \beta \in \mathbb{C}$ to write the qubit state as a superposition of excited/ground state as:

$$\left|\psi\right\rangle = \alpha\left|e\right\rangle + \beta\left|g\right\rangle \qquad \text{where, } |\alpha|^2 + |\beta|^2 = 1 \tag{8}$$

By getting ahead of QMT, we can introduce the probability to find a system in one of the possible eigenstates of an observable. Let us take the following example, what is the probability to find the energy E_n, defined by Eq. (6), from a system in $\left|e\right\rangle$ state? From a system in $\left|g\right\rangle$ state?

$$\mathbb{P}_e[E_n] = |\langle\phi_n|e\rangle|^2 \tag{9}$$
$$\mathbb{P}_g[E_n] = |\langle\phi_n|g\rangle|^2 \tag{10}$$

A physical state can be, by the concepts of quantum mechanics, a mixture of statistical probabilities [2]. Let us now consider back the superposition from Eq. (8), a physical mixture would be described by mixing states $\left|e\right\rangle, \left|g\right\rangle$ with weights $|\alpha|^2, |\beta|^2$:

$$\mathbb{P}_\psi[E_n] \stackrel{?}{=} |\alpha|^2\ \mathbb{P}_e[E_n] + |\beta|^2\ \mathbb{P}_g[E_n] \tag{11}$$

In fact, quantum mechanics gives the exact value of this probability:

$$\mathbb{P}_\psi[E_n] = |\langle\phi_n|\psi\rangle|^2 \tag{12}$$
$$= |\alpha|^2|\langle\phi_n|e\rangle|^2 + |\beta|^2|\langle\phi_n|g\rangle|^2 + 2\,\mathrm{Re}\left\{\alpha\beta^*\,\langle\phi_n|e\rangle\,\langle\phi_n|g\rangle^*\right\} \tag{13}$$

The mixture term appears but the impact of cross terms also appear in the real probability to find the superposition in one of the eigenstates. This notion of hidden correlations due to superposition has been highlighted by a new formalism, **density matrices**.
Basically, instead of a ket $\left|\psi\right\rangle$ we describe a system by a density matrix ρ which, for **pure** systems is simply:

$$\rho_{\mathrm{p}} = \left|\psi\right\rangle\left\langle\psi\right| \tag{14}$$

The density matrix is thus an operator of Hilbert space elements[5] $\rho : \mathcal{H} \to \mathcal{H}$. For a **statistical mixture**, thus non-pure, the density matrix in a basis $\{\phi_n\}_{n \in I_{\mathcal{H}}}$ takes the form:

$$\rho_{\text{sm}} = \sum_{n \in I_{\mathcal{H}}} \mathbb{P}_n \left|\phi_n\right\rangle\!\left\langle\phi_n\right| \tag{15}$$

The components of the matrix are precisely, on the diagonal, the probabilities to find the state of the system in one of the eigenstates of the observable. The off-diagonal components are called **coherences**, they represent interference effects between the eigenstates, they can appear when the system's state is a linear coherent superposition of these eigenstates [2].
The mathematical definition of a density matrix is in any case [2, 3]:

- ρ is hermitian $\rho^\dagger = \rho$
- ρ is semi-definite $\forall \left|\alpha\right\rangle \in \mathcal{H}, \quad \left\langle\alpha\right| \rho \left|\alpha\right\rangle \geq 0$
- ρ respects $\text{Tr}\ [\rho] = 1$

I.i.4⟩ Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathcal{L}(\mathcal{H})$. For pure states we can describe them using the Dirac formalism $\left|\psi\right\rangle \in \mathcal{H}$, or using the pure matrix density formalism $\rho_p = \left|\psi\right\rangle\!\left\langle\psi\right|$.

2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\left|a\right\rangle = a\left|a\right\rangle$.

4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}(a_n)$ (which depends on the spectrum of the observable).

5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement becomes the eigenstate $\left|\phi_n\right\rangle$.

6. The time evolution of the wavefunction $\left|\psi(t)\right\rangle$ is governed by the *Schrödinger* equation:
$$\frac{\mathrm{d}\left|\psi(t)\right\rangle}{\mathrm{d}t} = -\frac{i}{\hbar}\hat{H}(t)\left|\psi(t)\right\rangle$$
For density matrices, the evolution is driven by the *Liouville-von Neumann* equation[6]:
$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\left[H(t), \rho(t)\right] \equiv \mathcal{L}(t)\rho(t) \tag{16}$$

[5] ρ is also an operator of the dual, $\mathcal{H}^*$ because we find that $\left\langle\phi_j\right| \rho \propto \left\langle\phi_j\right|$.
We could even ague that $\rho \in \mathcal{L}(\mathcal{H})\ldots$

[6] $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is the commutator of $\hat{A}$ and $\hat{B}$

I.ii) Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. We recall that this description is ideal in the sense that the system is considered to be completely isolated from any interaction with its environment. However, we will introduce the OQS theory which develops a formalism to handle interaction of the system with its environment.
It is important to understand that the total system is a closed system, modelled by this relation on Hilbert spaces: $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. The total system T thus follows a unitary time evolution, but, it is not the case for the physical system of interest S. We therefore call the system S open quantum system.

In this theory, following the work of [3], we need to jump in a different representation of the equations: the interaction picture. In the *Schrödinger* picture, the states evolve with respect to time but the observables are constants $|\psi(t)\rangle = U(t, t_0)\,|\psi(t_0)\rangle$; whereas in the *Heisenberg* picture it is the opposite $A(t) = U^\dagger(t, t_0)A(t_0)U(t, t_0)$. The interaction picture is a mix of both, for further details please find Appendix A.

Interaction picture begins with the perturbation theory, where we split the Hamiltonian in two parts,

$$H_T = \overbrace{H_S + H_E}^{H_0} + H_I \tag{17}$$

$$U_T(t, t_0) = U_0(t, t_0)U_I(t, t_0) \tag{18}$$

We define the interaction Hamiltonian by,

$$E_n(t) = U_0^\dagger(t)E_nU_0(t) \tag{19}$$

$$H_I(t) = \sum_{n\in I_{\mathcal{H}}} \overbrace{|n\rangle\langle n|}^{A_n} \otimes E_n(t) \tag{20}$$

The main assumption states that observables of the system A_n commute with the free Hamiltonian and the total Hamiltonian $[H_0 + H_I, A] = [H_0, A] = 0$, they are thus constant of movement for the free/total systems. Also, the free Hamiltonian is constant in average $\frac{\mathrm{d}\langle H_S\rangle}{\mathrm{d}t}(t) = 0$.
We choose $H_E = \sum_n \hbar\omega_n(b_n^\dagger b_n)$ the harmonic oscillator model for the environment, where b_n is an annihilation operator of the environment. The following derivations let H_S free to pick, the interaction picture allows one to find the equation of the dynamic ...

I.iii) Quantum Measurement Theory (QMT)

II Continuously measured quantum systems – obtaining a Stochastic Master Equation (SME)

A Representations of quantum mechanics

A.a) Schrödinger and Heisenberg pictures

In the *Schrödinger* picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$ where, $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. The following demonstration can be made in the general case where the Hamiltonian is time dependent using Eq. (**??**).

If we consider the average value of an observable A (constant in this representation),

$$\begin{aligned}\langle A\rangle(t) &= \langle\psi(t)|A|\psi(t)\rangle \\ &= \langle\psi|\underbrace{U^\dagger(t)AU(t)}_{A(t)}|\psi\rangle\end{aligned} \tag{A. 1}$$

This naturally make appear the *Heisenberg* representation of observables.
We can link both representation with the following equation:

$$\begin{aligned}\frac{\mathrm{d}A(t)}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[U^\dagger(t)AU(t)\right] \\ &= \frac{\mathrm{d}U^\dagger(t)}{\mathrm{d}t}AU(t) + U^\dagger(t)\frac{\partial A}{\partial t}U(t) + \frac{\mathrm{d}U(t)}{\mathrm{d}t}AU(t) \\ &= \frac{i}{\hbar}U^\dagger(t)HAU(t) + U^\dagger(t)\frac{\partial A}{\partial t}U(t) - \frac{i}{\hbar}U^\dagger(t)AHU(t) \\ &= \frac{i}{\hbar}\left[U^\dagger HUU^\dagger AU - U^\dagger AUU^\dagger HU\right](t) + \frac{\partial A}{\partial t}(t) \\ &= \frac{i}{\hbar}\left[H(t)A(t) - A(t)H(t)\right] + \frac{\partial A}{\partial t}(t) \\ &= \frac{i}{\hbar}\left[H(t), A(t)\right] + \frac{\partial A}{\partial t}(t)\end{aligned} \tag{A. 2}$$

This link can even lead to the famous *Ehrenfest* theorem for quantum mechanics:

$$\frac{\mathrm{d}\langle A\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\langle[H(t), A(t)]\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle \tag{A. 3}$$

To summarize each equation for time evolution in both pictures:

$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \frac{i}{\hbar}H(t)\,|\psi\rangle \qquad (\textit{Schrödinger})$$

$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = \frac{i}{\hbar}[H(t), \rho] \qquad (\textit{Liouville-von Neumann})$$

$$\frac{\mathrm{d}A(t)}{\mathrm{d}t} = \frac{i}{\hbar}\left[H(t), A(t)\right] + \frac{\partial A}{\partial t}(t) \qquad (\textit{Heisenberg})$$

A.b) Interaction picture

References

[1] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics, Vol. III, Chapter 1: Quantum Behavior*. California Institute of Technology, accessed 2026-04-02. 1965. URL: https://www.feynmanlectures.caltech.edu/III_01.html.

[2] Claude Cohen-Tannoudji, Bernard Diu, and Franck Laloë. *Mécanique quantique*. fr. Companyédition CNRS/EDP Sciences, Sept. 2018.

[3] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University PressOxford, Jan. 2007. ISBN: 9780191706349. DOI: 10.1093/acprof:oso/9780199213900.001.0001. URL: http://dx.doi.org/10.1093/acprof:oso/9780199213900.001.0001.

[4] Kurt Jacobs. *Quantum Measurement Theory and its Applications*. en. Cambridge, England: Cambridge University Press, Aug. 2014.

[5] Francesco Albarelli and Marco G. Genoni. "A pedagogical introduction to continuously monitored quantum systems and measurement-based feedback". In: *Physics Letters A* 494 (Jan. 2024), p. 129260. ISSN: 0375-9601. DOI: 10.1016/j.physleta.2023.129260. URL: http://dx.doi.org/10.1016/j.physleta.2023.129260.